SUMMARY

Task 1 — Data Loading & Exploration:

We loaded the dataset and explored its structure. The target variable was **price**, while features included area, bedrooms, bathrooms, stories, parking, furnishing status, and amenities. The dataset contained no major missing values after inspection.

Task 2 — Data Cleaning:

We handled missing values by filling numeric columns with medians and categorical columns with modes. Duplicate rows were removed, and categorical variables such as *yes/no* fields and furnishing status were converted into numeric form using one-hot encoding.

Task 3 — Model Building:

Two models were trained: **Linear Regression** and **Random Forest Regressor**. Random Forest achieved better accuracy, with lower MAE and RMSE and a higher R^2 score, showing it captured complex relationships more effectively than Linear Regression.

Task 4 — Visualization:

We created three charts:

- A histogram showing the distribution of house prices.
- A correlation heatmap highlighting strong relationships (area, bedrooms, bathrooms, parking) with price.
- An actual vs predicted scatter plot for Random Forest, showing close alignment between predictions and true values.

Task 5 — Insights & Summary:

Key features influencing price were **area, number of rooms, parking spaces, and furnishing status**. The Random Forest model was fairly accurate in plain terms, predicting prices close to actual values. A surprising insight was the strong impact of amenities like air conditioning and furnishing compared to some structural features. For real estate businesses, the recommendation is to emphasize modern amenities and furnishing quality when marketing properties, as these significantly boost perceived value and selling price.